AP CALCULUS AB

Unit 1 • Lesson 1.5 practice • Answer key

Lesson 1.5 • Determining Limits Using Algebraic Properties of Limits

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$\lim_{x \to 2} \left(3x^2 - x\right)$ is

(A) 10

(B) 12

(C) 14

(D) 8

Solution

Answer: (A). Direct substitution: $3(4) - 2 = 10$.

Question 2

NO CALCULATOR

If $\lim_{x \to c} f(x) = 4$ and $\lim_{x \to c} g(x) = -1$, then $\lim_{x \to c} \left(f(x)g(x) + 3\right)$ is

(A) 7

(B) -1

(C) 3

(D) -4

Solution

Answer: (B). $4 \cdot (-1) + 3 = -1$.

Question 3

NO CALCULATOR

If $\lim_{x \to c} f(x) = 6$ and $\lim_{x \to c} g(x) = 2$, then $\lim_{x \to c} \dfrac{f(x)}{g(x)}$ is

(A) 12

(B) 4

(C) 3

(D) $\dfrac{1}{3}$

Solution

Answer: (C). The denominator's limit is not zero, so the limit is $\frac{6}{2} = 3$.

Question 4

NO CALCULATOR

$\lim_{x \to \pi} \cos x$ is

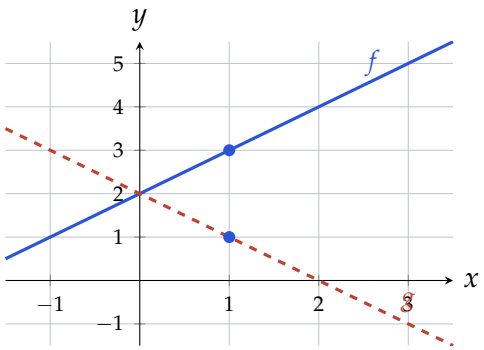
(A) 0
(B) 1
(C) π
(D) -1

Solution
Answer: (D). $\cos x$ is continuous, so the limit is $\cos \pi = -1$.

Question 5 (graph)

NO CALCULATOR

The graphs of the lines f (solid) and g (dashed) are shown. What is $\lim_{x \to 1} f(x)\, g(x)$?



(A) 4
(B) 2
(C) 1
(D) 3

Solution
Answer: (D). From the graph, $\lim_{x \to 1} f(x) = 3$ and $\lim_{x \to 1} g(x) = 1$. By the product property the limit is $3 \cdot 1 = 3$.

Question 6 (Free response)

NO CALCULATOR

$\lim_{x \to 1} f(x) = 3$, $\lim_{x \to 1} g(x) = -2$ and $h(x) = x^2 + 1$. Find each limit, showing which properties you use.

(a) $\lim_{x \to 1} \big(2f(x) - g(x)\big)$ (b) $\lim_{x \to 1} f(x)h(x)$
(c) $\lim_{x \to 1} \sqrt{f(x) + 6}$ (d) $\lim_{x \to 1} \big(g(x)\big)^3$

Solution
(a) Constant multiple and difference: $2(3) - (-2) = 8$.
(b) $\lim_{x \to 1} h(x) = 2$ by substitution; product: $3 \cdot 2 = 6$.
(c) Sum, then the square root (continuous): $\sqrt{3 + 6} = 3$.
(d) Power: $(-2)^3 = -8$.